

Adhithi Mudaliyar

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EDUCATION

University of South Florida

Bachelor of Science in *Computer Science & Economics*

Grad: May 2028

GPA: 3.80/4.00

- **Relevant Coursework:** Data Structures & Algorithms, Probability & Statistics, Machine Learning, NLP

TECHNICAL SKILLS

AI-Assisted Development: Claude Code, Codex, GitHub Copilot, CLAUDE.md, spec templates, AI code review

Languages: Python, Java, C/C++, JavaScript, SQL

Software Engineering: Git/GitHub, Azure DevOps, pytest, unit/integration testing, REST APIs, Docker, Linux, Agile

Data & Cloud: Pandas, NumPy, PostgreSQL, Spark, Airflow, Databricks, AWS (S3, SageMaker)

ML & NLP: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, retrieval-augmented generation (RAG), SHAP

EXPERIENCE

Research Assistant

Sept. 2026 – Present

USF Health, Center for Advanced Medical Learning and Simulation

Tampa, FL

- Validate experimental logging and simulation behavior through **end-to-end testing** of an **XR medical-training platform**, catching data-integrity issues before they reach analysis.
- Standardize simulation errors, completion times, and device conditions into analysis-ready **Python pipelines**, replacing per-study manual cleanup with one documented, reproducible method for research collaborators.

Student Researcher

Apr. 2026 – Present

University of South Florida, Prof. Yu Sun

Tampa, FL

- Resolved incompatible **ADNI** and **HABS-HD** schemas blocking cross-cohort analysis; built Pandas pipelines harmonizing a **32-feature multimodal dataset** so **5 research teams** worked from one analysis-ready source.
- Delivered scikit-learn classification workflows with reproducible 5-fold cross-validation and error analysis, reaching **81.5% and 97.2% held-out accuracy**.

Software/Firmware Engineer Intern

May 2026 – Aug. 2026

Elevate Healthcare

Sarasota, FL

- Built Python/Linux test tooling and embedded **C/C++** to integrate and validate **4 healthcare simulation subsystems** in a **regulated medical-device environment**.
- Directed **Claude Code** and **GitHub Copilot** to navigate unfamiliar embedded codebases and draft test harnesses, critically reviewing every AI-generated change against subsystem specifications before merge.
- Documented reproducible integration tests and handoff procedures adopted by the engineering **team of 7**; tracked defects and validated device behavior across **Agile sprints** using Git and **Azure DevOps**.

PROJECTS

MedLens | Python, SQL, Airflow, Spark/Databricks, AWS S3, XGBoost, pytest, Claude Code

Sept. 2026

- Built end to end with **Claude Code**; authored a **CLAUDE.md** repository context file and a reusable task-spec template (objective, constraints, acceptance criteria), decomposed the build into staged tasks, and reviewed every diff, re-scoping instructions whenever output missed a criterion.
- Shipped Airflow/S3 pipelines processing **5 million CMS Part D records** across 3 annual releases, with **pytest data-validation tests** gating every stage against schema and range regressions.
- Forecast 1-year-ahead drug spend with XGBoost on time-based holdouts, beating a last-year baseline by **10% MAE**; surfaced results through Power BI and Streamlit dashboards.

Diabetes Risk Intelligence System | Python, FastAPI, Docker, AWS SageMaker, pytest, XGBoost

May 2025

- Deployed a containerized **FastAPI REST service** on SageMaker serving calibrated risk predictions (**0.89 ROC-AUC**) with SHAP explanations in **under 120 ms**.
- Wrote **pytest** suites validating the retraining pipeline and API responses to protect against regressions.

ToxiScope | Python, PyTorch, Hugging Face, RAG, Streamlit

Jan. 2026

- Built a PubMed **RAG system** (BM25 + dense retrieval) and active-learning toxicity models that matched baseline ROC-AUC with **40% fewer labeled compounds**; verified citations across 50 queries.

LEADERSHIP

Technical Lead

Apr. 2025 – Present

Association for Computing Machinery, USF Chapter

Tampa, FL

- Lead a 10-member technical team delivering AI and software workshops to **50+ students weekly**; mentor on debugging, testing, and Git-based collaboration.